



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: MCA
Semester: II
Name of the Paper: Machine Learning-1
Paper Code: -TMC 211
Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1.

(10 Marks) [C.O - I]

a.: Explain the followings.

- i) Entropy ii) information gain iii) Classification iv) Machine V) Accuracy of a model

OR

b. Distinguish between supervised learning and Reinforcement learning. Illustrate with an example. Discuss any four examples of machine learning applications.

[C.O - I]

Q2.

(10 Marks)

[C.O - 2]

a. Give - ID3 algorithm for construction of decision tree.

OR

b. Create Confusion matrix and find out Accuracy, Precision and F-1 Score for prediction machine learning model.

[C.O - 2]

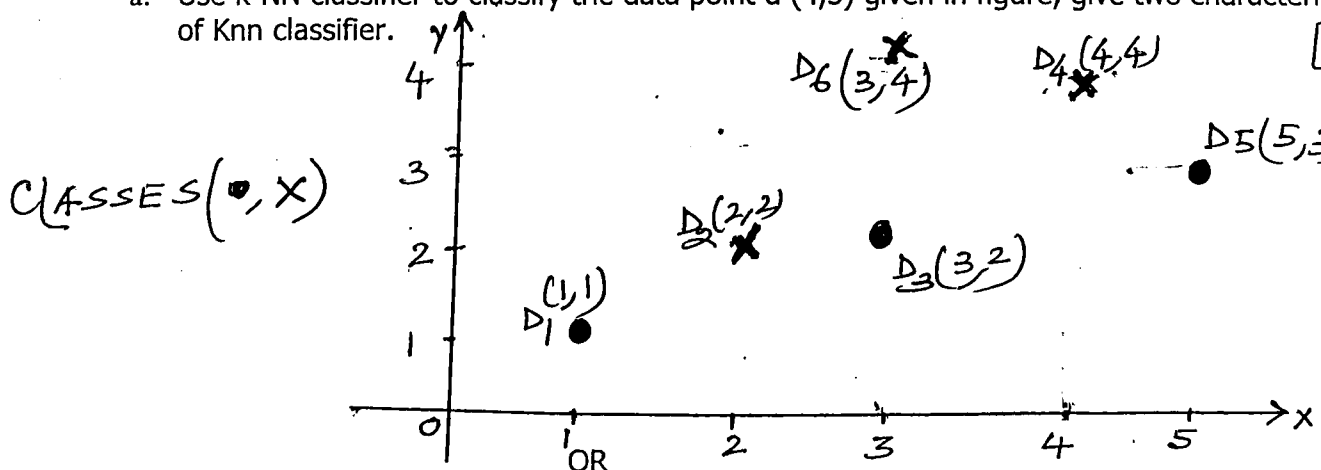
ACTUAL	Y	Y	N	Y	N	Y	Y	N	Y	Y
PREDICTED	Y	Y	Y	N	N	Y	N	N	Y	N

Q3.

(10 Marks)

- a. Use k-NN classifier to classify the data point u (4,5) given in figure, give two characteristic of Knn classifier.

[C.O - 3]





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b.: Write short notes on any one in brief

[C.O-2]

- i) Numerical python (NumPy)
- ii) Logistic Regression

Q4. (10 Marks)

a. What is cross validation? Explain the various methods to perform cross validation

[C.O-4]

OR

b. What is List and tuple in Python? How the slicing, addition and deletion operation performed on list explain with example? [C.O-2]

Q5. (10 Marks)

a. Discuss Kernel trick in SVM with help of an example.

[C.O-3]

OR

b.: Fit a line this data (linear regression) and find value of y for x=11.

[C.O-2]

x	:	1	2	4	6	8	9
y	:	3	4	7	4	10	12